artefacts of the working cannula on the CBCT the comparison between MChet simulations on CBCT and pre-op CT showed differences up to 50% in dose. The maximum dose in the spinal cord (distance of 11mm from applicator tip) was 11Gy for the MCwater and 7.5Gy for the MChet simulations on pre-op CT.

Conclusion

Precision IORT using a combination of intraoperative image guidance and treatment planning improves the accuracy of IORT. However, the current set-up is limited by CT artefacts. Fusing an intraoperative CBCT with a pre-op CT allows the combination of an accurate dose calculation with the knowledge of the correct source/ applicator position. This method can also be used for pre-operative treatment planning followed by image guided surgery.

Proffered Papers: Eye/GYN

OC-0363 Ruthenium-106 brachytherapy for iris and choroidal body melanomas

F.P. Peters[1], M. Marinkovic[2], N. Horeweg[1], M.S. Laman[1], J.C. Bleeker[2], M. Ketelaars[1], G.P.M. Luyten[2], C.L. Creutzberg[1]
[1]*Leiden University Medical Center LUMC, Department of Radiotherapy, Leiden, The Netherlands*

[2]*Leiden University Medical Center LUMC, Department of Ophtalmology, Leiden, The Netherlands*

Purpose or Objective

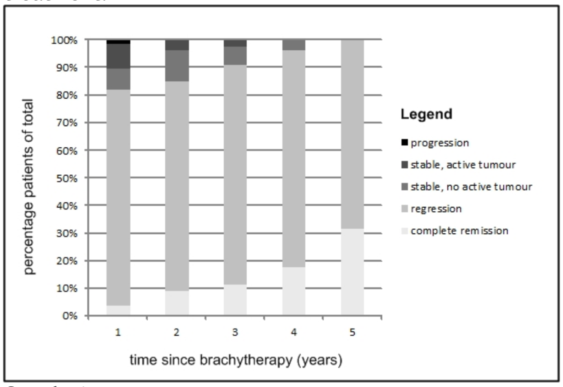
Uveal melanoma is a malignant neoplasm that arises from the neuro-ectodermal melanocytes within the choroid, ciliary body or iris. Ninety percent of uveal melanomas are choroidal melanomas (CM), only six percent originates in the ciliary body and 4% in the iris. Eye-conserving treatment of small to intermediate-sized CM by Ruthenium-106 brachytherapy (Ru106) yields a 95% 5-year local control rate (Marinkovic et al., Eur J Cancer, 2016). Disadvantage of this treatment is that visual acuity decreases to <0.33 in 25% of the patients. Small to intermediate-sized iris melanomas (IM) and choroidal body melanomas (CBM) are also treated by Ru106. As the localisation of the tumour and the organs at risk in IM and CBM are different from those in CM, treatment effectiveness and complications may also differ. This study was conducted to assess outcomes of Ru106 as eye-conserving treatment of IM and CBM in terms of local control, metastasis, survival, eye preservation, treatment toxicity and visual outcomes.

Material and Methods

Data was collected on 88 consecutive patients who were treated for IM or CBM from 2006 to 2016. Minimal radiation dose was 120-130Gy; specified at the depth of the tumour base (for IM) or tumour apex (for CBM); provided a maximal corneal dose <500-600Gy, scleral dose <1000Gy and application time <5-6 days. Primary outcome of this study was local control. Secondary outcomes were metastasis, melanoma-related death, eye preservation, treatment complications and post-treatment visual acuity. Durations were calculated using Kaplan-Meier's methodology, risk factors were assessed using a Cox proportional hazards model.

Results

Total median follow-up was 36 months (range: 3-115). Of 88 patients, 58 (65.9%) were diagnosed with IM and 30 (34.1%) with CBM. At diagnosis, CBM were larger and more advanced than IM. Figure 1 presents the results of the yearly local site evaluation after treatment. Hence, tumour regression evolved steadily over the years, with >80% already showing regression after one year. Local control rate at the end of follow-up of all tumours was 98.9%. Metastases were diagnosed in 1.1% of the patients; no deaths due to melanoma occurred during follow-up. Eye preservation rate during follow-up was 97.7%. Treatment-related toxicities were observed in 80.7% of the patients, however most toxicities were mild and transient. Worsening of pre-existing or new cataract was observed in 51.1%; 64.4% of these patients underwent cataract extraction after brachytherapy. Further, dry eyes (29.5%) and glaucoma were (20.5%) commonly observed toxicities. Visual acuity was not affected by Ru106 brachytherapy, with only 2.3% having a visual acuity <0.33 (no useful vision) at follow-up, compared to 13.6% before treatment.



Conclusion

Ru106 for IM and CBM yielded excellent local control rate of 98.9% and 97.7% eye preservation. Treatment toxicities were common, but mostly mild and transient. Moreover, Ru106 did not affect visual acuity.

OC-0364 Nomogram for predicting maculopathy in patients treated with Ru106 brachytherapy for uveal melanoma

L. Tagliaferri[1], A. Larichiuta[1], M. Pagliara[2], C. Masciocchi[3], J. Lenkowicz[3], L. Azario[4], R. Autorino[1], M.A. Gambacorta[1], V. Valentini[3], M.A. Blasi[2]
[1]*Fondazione Policlinico Universitario A. Gemelli, Dipartimento di Radioterapia Oncologica - Gemelli ART, Roma, Italy*
[2]*Fondazione Policlinico Universitario A. Gemelli, Dipartimento di Oftalmologia, Roma, Italy*
[3]*Università Cattolica del Sacro Cuore, Dipartimento di Radioterapia Oncologica - Gemelli ART, Roma, Italy*
[4]*Fondazione Policlinico Universitario A. Gemelli, Unità Complessa di Fisica Sanitaria, Roma, Italy*

Purpose or Objective

Plaque brachytherapy (BT) as a practicable alternative to enucleation for the treatment of medium-sized choroidal melanomas. However, the BT is not free from local toxicity. The aim of this study was to develop a predictive model for maculopathy occurrence after ruthenium-106 plaque brachytherapy in uveal melanoma.

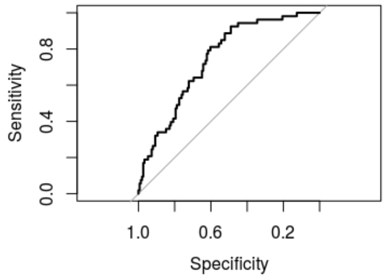
Material and Methods

Patients from institutional database with choroidal melanoma treated with ruthenium-106 plaque from December 2006 to December 2014 were selected. Inclusion criteria were: dome-shaped melanoma, distance to the Fovea > 1.5 mm and tumor thickness > 1 mm and < 5mm. In each case, the prescribed dose was 100 Gy at tumor apex. Factors analyzed were sex, age, diabetes, tumor size (volume, area, largest basal diameter and apical height), plaque types, distance to fovea, presence of exudative detachment, presence of drusen, presence of orange pigments, radiation dose to the fovea and sclera. Univariate and multivariate Cox proportional hazards were used to define the impact of baseline patient factors on the occurrence of the maculopathy. A p-value <= 0.05 was considered significant. Kaplan-Meier curves were used to estimate freedom from the occurrence of the maculopathy. The model performance was evaluated with internal validation using Area Under the ROC Curve (AUC) and calibration with Hosmer-Lemeshow test.

Results

Two hundred and five patients with a median age of 68 (range: 17-92 years) were considered for this analysis. The median follow-up was 41 months. Of 205 patients, 92% were alive. Maculopathy was found in 53 patients (25.8%) after the treatment. Distance to fovea was the main prognostic factor of the predictive model (hazard ratio [HR] of 0.813 [0.75-0.87] p = 3.45e-08). Diabetes (hazard radio [HR] of 2.31 [1.14-4.66], p = 0.019), and tumor volume (hazard radio [HR] of 19.08 [2.06-175.88], p = 0.0093) affected the prediction of maculopathy. The prediction model developed can predict events of maculopathy at 3 years with an AUC of 0.74 (figure 1). The calibration showed no statistical difference between actual and predicted maculopathy (p=0.22).

Figure1: Model's ROC. AUC = 0.745



Conclusion

Our maculopathy prognostication model, along with its nomogram, could be a tool for predicting the occurrence of maculopathy at 3 years after treatment. Furthermore, this analysis revealed that tumor volume, distance to the fovea and diabetes can help to predict maculopathy at 3 years after treatment: a predictive model (coefficients and nomogram) is provided and good performance obtained encourage further investigations along this direction.

OC-0365 Dose contribution to pelvic nodes of image-guided adaptive brachytherapy in cervical cancer

W. Bacorro[1,2], I. Dumas[3], A. Levy[2], E. Rivin del Campo[2], C.H. Canova[2], T. Felefly[2], A. Huertas[2], F. Marsolat[3], P. Maroun[2], C. Haie-Meder[2], C. Chargari[2], R. Mazeron[2]
[1]*Benavides Cancer Institute- UST Hospital, Radiation Oncology, Manila, Philippines*
[2]*Institute Gustave Roussy, Radiation Oncology, Villejuif, France*
[3]*Institute Gustave Roussy, Medical Physics, Villejuif, France*

Purpose or Objective

The use of simultaneous integrated boost (SIB) to pathologic pelvic nodes in the treatment of cervical cancer requires integrating in the IMRT plan the contribution of brachytherapy. This study aims to report the BT-delivered doses to pelvic pathologic nodes and to propose SIB dose-fractionation regimens.

Material and Methods

Patients with locally advanced cervical cancer comprising pelvic nodal involvement and treated with chemoradiation followed by image-guided adaptive pulsed-dose rate BT were included. The pathologic nodes were delineated to report the brachytherapy contribution but without planning aims. D_{100}, D_{98}, D_{90} and D_{50} were reported and converted to 2-Gy equivalents (EQD2), using the linear quadratic model with an α/β of 10 Gy.

Results

Ninety-one patients were identified, allowing the evaluation of dose delivery in 226 adenopathies. The majority of the studied nodes were located in the external iliac (48%), common iliac (25%), and internal iliac (16%) regions. The EQD2 contribution was 3.6±2.2 Gy, 4.1±1.6, 4.4±3.3, and 5.2±3.9 Gy for the D100, D98, D90, and D50, respectively. The EQD2 D_{98} values were 4.4±1.9 Gy, 5.4±3.1 Gy, 4.3±2.1 Gy for obturator, internal iliac and external iliac nodes respectively, and 2.8±2.5 Gy for the common iliac. Whereas no significant difference was observed between the brachytherapy contributions of external and internal iliac nodes, the doses delivered in common iliac adenopathies were significantly lower (p<0.001).

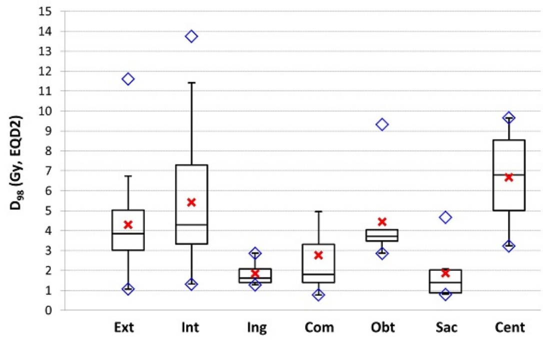


Figure: Descriptive statistics of D98 of pathologic nodes according to regions.
Ext: external iliac, Int: internal iliac, Ing: inguinal, Com: common iliac, Obt: obturator, Sac: presacral, Cent: central (pararactal or parametrial). Red cross: mean value, blue diamond: minimal and maximal values, lower limit of the box: first quartile, upper limit of the box: third quartile, central horizontal bar: median, whiskers: from minimal value to 1.5 x box length.

Thus, to deliver a cumulative EQD2 ≥60 Gy to pathologic nodes accounting a pelvic external beam radiation dose of 45 Gy in 25 fractions (44.3 in EQD2) and these estimations, we propose nodal SIB of 2.2 Gy x 25 (55 Gy, 55.9 in EQD2) in the obturator, external and internal iliac nodes, 2.3 Gy x 25 (57.5 Gy, 58.9 in EQD2) in the common iliac nodes, and 2.4 Gy x 25 (60 Gy, 62 Gy in EQD210) in the para-aortic nodes (where the BT contribution can be considered as negligible).

Conclusion

The contribution of brachytherapy to the treatment of pelvic nodes is significant: around 5 Gy in the obturator, internal iliac, and external iliac areas and 2.5 Gy in the common iliac, allowing the use of simultaneous integrated boost. However, important individual variations have been observed and evaluation of the genuine individual brachytherapy contribution is recommended.

OC-0366 Cervical cancer with bladder invasion: outcomes and vesicovaginal fistula prognostic factors

R. Sun[1], R. Mazeron[1], I. Koubaa[2], I. Dumas[3], C. Baratiny[1], F. Monnot[1], P. Maroun[1], E. Deutsch[1], P. Morice[4], C. Haie-Meder[1], C. Chargari[1]
[1]*Gustave Roussy, Radiation oncology, Villejuif, France*
[2]*Gustave Roussy, Radiology, Villejuif, France*
[3]*Gustave Roussy, Medical physics, Villejuif, France*
[4]*Gustave Roussy, Surgery, Villejuif, France*

Purpose or Objective

Although brachytherapy (BT) is a mainstay of the treatment of locally advanced cervical cancer, there are only scarce data on its efficiency in cervical cancer with bladder invasion. The aims were to report the treatment outcomes in this particular situation, as well as vesicovaginal fistula (VVF) incidence and its prognostic factors.

Material and Methods

Consecutive patients with locally advanced cervical cancer and bladder invasion treated in our institution from 1989 to 2015 were identified. Demographic and tumor features, treatment characteristics, VVF rate, progression-free survival (PFS), local control rate (LCR), and overall survival (OS) were reviewed. Baseline